

Cybersecurity Training Diary

Day 16: Incident Response & SOC Fundamentals

1. Topics Covered

✓ Incident Response Lifecycle

```
graph LR
    A[Preparation] --> B[Identification]
    B --> C[Containment]
    C --> D[Eradication]
    D --> E[Recovery]
    E --> F[Lessons Learned]
```

✓ SOC Analyst Tools

Tool Type	Purpose	Examples
SIEM	Centralized log analysis	Splunk, Sentinel, QRadar
EDR	Endpoint detection & response	CrowdStrike, Carbon Black
Network Monitoring	Traffic analysis	Zeek, Wireshark, Suricata

✓ Common Alert Types

- **Brute Force Attempts** (Multiple failed logins)
- **Malware Signatures** (Known hash matches)
- **Data Exfiltration** (Unusual outbound traffic)
- **Policy Violations** (Unauthorized USB usage)

2. Key Takeaways

- ✓ **Mean Time to Detect (MTTD)** is just as important as Mean Time to Respond (MTTR)
- ✓ **False positives waste resources** - tune your SIEM rules regularly
- ✓ **Document everything** - chain of custody matters for legal cases

3. Practical Exercises

♦ Exercise 1: Analyze SIEM Alerts

```
# Sample Splunk Query for Brute Force
index=security sourcetype=linux_secure "Failed password"
| stats count by src_ip, user
| where count > 5
```

♦ Exercise 2: Memory Forensics Basics

```
# Using Volatility (Linux)
vol.py -f memory.dump windows.pslist
vol.py -f memory.dump windows.malfind
```

4. Questions for Reflection

- ? Why would an organization choose a managed SOC vs in-house?
- ? How could AI/ML improve threat detection?